

IMPORTIING LIBRARIES

```
In [4]: import pandas as pd
import numpy as numpy
```

LOAD DATA

```
In [5]: df = pd.read_csv ( 'C:/Users/Nancy/Desktop/School Of Statisticians Projects/Pyth
```

```
In [6]: df.head()
```

Out[6]:

	review_id	customer_name	raw_review	product	rating	date	cleaned_review
0	1	Craig Mckenzie	Little understand try sell citizen create no m...	Toaster	3	2025-05-31	NaN
1	2	Brian Coleman	Hair emplo\ (yee on.	Smartwatch	3	2025-06-02	NaN
2	3	Jeremy Price	Five p~ {#layer rest himself.	Smartwatch	4	2025-05-01	NaN
3	4	Jose Roth	Off try shake care health nation identify mode...	Phone Case	5	2025-05-17	NaN
4	5	Allen Sims	New m~,(! @!::~y sport management outside cente...	Headphones	3	2025-05-19	NaN

```
In [7]: df.describe()
```

Out[7]:

	review_id	rating	cleaned_review	review_length
count	5000.000000	5000.000000	0.0	0.0
mean	2500.500000	2.991200	NaN	NaN
std	1443.520003	1.402967	NaN	NaN
min	1.000000	1.000000	NaN	NaN
25%	1250.750000	2.000000	NaN	NaN
50%	2500.500000	3.000000	NaN	NaN
75%	3750.250000	4.000000	NaN	NaN
max	5000.000000	5.000000	NaN	NaN

In [8]: `df.isnull().sum()`

Out[8]:

review_id	0
customer_name	0
raw_review	0
product	0
rating	0
date	0
cleaned_review	5000
review_length	5000

dtype: int64

UNDERSTANDING DATA

1. What are the main columns in the dataset and what does each represent?

In [9]: `df.columns`

Out[9]: Index(['review_id', 'customer_name', 'raw_review', 'product', 'rating', 'date', 'cleaned_review', 'review_length'], dtype='object')

product: Name or ID of the product being reviewed.

rating: Numerical score given by the user (e.g., 1–5).

raw_review: Original text written by the user.

Other possible columns: user_id, timestamp, etc.

2. What are some examples of messy elements in the raw_review column?

In [35]: `pd.set_option("display.max_rows", None)
df["raw_review"].head(10)`

```
Out[35]: 0 Little understand try sell citizen create no m...
        1 Hair emplo\(\yee on.
        2 Five p~{ }#layer rest himself.
        3 Off try shake care health nation identify mode...
        4 New m~,(!@!~y sport management outside cente...
        5 He happy red 🍌:$.former rate manage.
        6 Entire least site black their line believe eve...
        7 Per than final suggest thro'{ughout event writ...
        8 Little interview between evening avoid shoulde...
        9 Money |&"😞him night way.
Name: raw_review, dtype: object
```

Emojis: 😊👍😡

Special characters: @#%&*!\$()

Numbers: "5 out of 10", "Rated 4"

Inconsistent casing: "GREAT PRODUCT", "bad service"

Extra whitespace: " Too slow "

3. How many unique products are reviewed in the dataset?

```
In [11]: df["product"].unique()
```

```
Out[11]: array(['Toaster', 'Smartwatch', 'Phone Case', 'Headphones', 'Shoes',
               'Blender', 'Mug'], dtype=object)
```

CLEANING TASKS

4. How can you remove all special characters, emojis, and numbers from the raw_review column using Python?

```
In [12]: pip install emoji
```

Requirement already satisfied: emoji in c:\users\nancy\anaconda3\lib\site-packages (2.14.1)

Note: you may need to restart the kernel to use updated packages.

```
In [13]: import re
import emoji
```

```
In [14]: # Function to clean each review
def clean_review(text):
    # Remove emojis
    text = emoji.replace_emoji(text, replace='')
    # Remove special characters and numbers
    text = re.sub(r'^A-Za-z\s', '', text)
    # Lowercase and strip extra spaces
    text = ' '.join(text.lower().split())
    return text
```

```
In [15]: # Apply cleaning
df["cleaned_review"] = df["raw_review"].apply(clean_review)
```

In [16]: `df.head()`

Out[16]:

	review_id	customer_name	raw_review	product	rating	date	cleaned_review
0	1	Craig Mckenzie	Little understand try sell citizen create no m...	Toaster	3	2025-05-31	little understand try sell citizen create no m...
1	2	Brian Coleman	Hair emplo\ (yee on.	Smartwatch	3	2025-06-02	hair employee on
2	3	Jeremy Price	Five p~ {}#layer rest himself.	Smartwatch	4	2025-05-01	five player rest himself
3	4	Jose Roth	Off try shake care health nation identify mode...	Phone Case	5	2025-05-17	off try shake care health nation identify mode...
4	5	Allen Sims	New m~,(! @!::~y sport management outside cente...	Headphones	3	2025-05-19	new my sport management outside center age opt...

5. Write a function in Python that strips extra spaces and lowercases the review text.

In [17]:

```
def tidy_text(text):
    return ' '.join(text.lower().split())

df['cleaned_review'] = df['cleaned_review'].apply(tidy_text)
```

In [18]: `df.head()`

Out[18]:

	review_id	customer_name	raw_review	product	rating	date	cleaned_review
0	1	Craig Mckenzie	Little understand try sell citizen create no m...	Toaster	3	2025-05-31	little understand try sell citizen create no m...
1	2	Brian Coleman	Hair emplo\ (yee on.	Smartwatch	3	2025-06-02	hair employee on
2	3	Jeremy Price	Five p~ {}#layer rest himself.	Smartwatch	4	2025-05-01	five player rest himself
3	4	Jose Roth	Off try shake care health nation identify mode...	Phone Case	5	2025-05-17	off try shake care health nation identify mode...
4	5	Allen Sims	New m~,(! 🗑@!::~y sport management outside cente...	Headphones	3	2025-05-19	new my sport management outside center age opt...



Input Handling: Checks if the input is a string; returns an empty string for None or non-string inputs.

Space Normalization: Uses `split()` to break the string into words (removing extra spaces) and `join()` to combine them with a single space.

Lowercase Conversion: Converts the text to lowercase using `lower()`.

6. What is the difference between `raw_review` and `cleaned_review` after text cleaning? Illustrate with 3 examples.

```
In [19]: def clean_review_text(raw_review):
  if not isinstance(raw_review, str):
      return ""
  cleaned = ' '.join(raw_review.split()).lower()
  return cleaned
```

Example 1 `raw_review`: " This is AMAZING!!! "

`cleaned_review`: "this is amazing!!!"

Explanation: Removed leading/trailing spaces and extra spaces between words.

Converted "This", "AMAZING" to lowercase.

Preserved "!!!" since the function doesn't remove punctuation.

Example 2 raw_review: "POOR Service... @\$123"

cleaned_review: "poor service... @\$123"

Explanation: Removed extra spaces between "Service..." and "@#\$123".

Converted "POOR", "Service" to lowercase.

Preserved "...", "@#\$", and "123" since the function doesn't remove special characters or numbers.

Example 3 raw_review: " Great Product 2023 "

cleaned_review: "great product 2023"

Explanation: Removed leading/trailing spaces and extra spaces between words.

Converted "Great", "Product" to lowercase.

Preserved "" and "2023" since the function doesn't remove emojis or numbers

CREATING NEW COLUMNS

7. How would you compute the number of words in each review and store it in a new column review_length?

```
In [20]: len("raw_review")
```

```
Out[20]: 10
```

```
In [21]: df['review_length'] = df['raw_review'].str.split().str.len()
```

```
In [22]: df.head()
```

Out[22]:

	review_id	customer_name	raw_review	product	rating	date	cleaned_review
0	1	Craig Mckenzie	Little understand try sell citizen create no m...	Toaster	3	2025-05-31	little understand try sell citizen create no m...
1	2	Brian Coleman	Hair emplo\ (yee on.	Smartwatch	3	2025-06-02	hair employee on
2	3	Jeremy Price	Five p~ {}#layer rest himself.	Smartwatch	4	2025-05-01	five player rest himself
3	4	Jose Roth	Off try shake care health nation identify mode...	Phone Case	5	2025-05-17	off try shake care health nation identify mode...
4	5	Allen Sims	New m~,(! 🗑@!::~y sport management outside cente...	Headphones	3	2025-05-19	new my sport management outside center age opt...

8. Categorize Reviews into 'Positive', 'Neutral', or 'Negative' Using the rating Column

```
In [23]: df["sentiment"] = df["rating"].apply(lambda x: "Positive" if x >= 4 else ("Neutr
```

```
In [24]: df.head()
```

Out[24]:

	review_id	customer_name	raw_review	product	rating	date	cleaned_review
0	1	Craig Mckenzie	Little understand try sell citizen create no m...	Toaster	3	2025-05-31	little understand try sell citizen create no m...
1	2	Brian Coleman	Hair emplo\ (yee on.	Smartwatch	3	2025-06-02	hair employee on
2	3	Jeremy Price	Five p~ {}#layer rest himself.	Smartwatch	4	2025-05-01	five player rest himself
3	4	Jose Roth	Off try shake care health nation identify mode...	Phone Case	5	2025-05-17	off try shake care health nation identify mode...
4	5	Allen Sims	New m~,(! @!::~y sport management outside cente...	Headphones	3	2025-05-19	new my sport management outside center age opt...

9. Which product received the most negative reviews (rating ≤ 2)?

```
In [25]: df[df["sentiment"]=="Negative"]["product"].unique()
```

```
Out[25]: array(['Toaster', 'Smartwatch', 'Shoes', 'Blender', 'Phone Case', 'Mug',  
              'Headphones'], dtype=object)
```

SUMMARY AND INSIGHT

11. What's the average review length for positive vs negative reviews?

```
In [26]: average_review_length= df[df["sentiment"].isin(["Positive","Negative"])].groupby  
print("Average Review Length:\n", average_review_length)
```

Average Review Length:

sentiment

Negative 9.641902

Positive 9.531504

Name: review_length, dtype: float64

12. Plot a bar chart showing the number of reviews per product. What insights can you draw?

```
In [27]: pip install seaborn
```

Requirement already satisfied: seaborn in c:\users\nancy\anaconda3\lib\site-packages (0.13.2)

Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\nancy\anaconda3\lib\site-packages (from seaborn) (1.26.4)

Requirement already satisfied: pandas>=1.2 in c:\users\nancy\anaconda3\lib\site-packages (from seaborn) (2.2.2)

Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\nancy\anaconda3\lib\site-packages (from seaborn) (3.8.4)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\nancy\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.2.0)

Requirement already satisfied: cycler>=0.10 in c:\users\nancy\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\nancy\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.51.0)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\nancy\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.4)

Requirement already satisfied: packaging>=20.0 in c:\users\nancy\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (23.2)

Requirement already satisfied: pillow>=8 in c:\users\nancy\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (10.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\nancy\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.0.9)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\nancy\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\users\nancy\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2024.1)

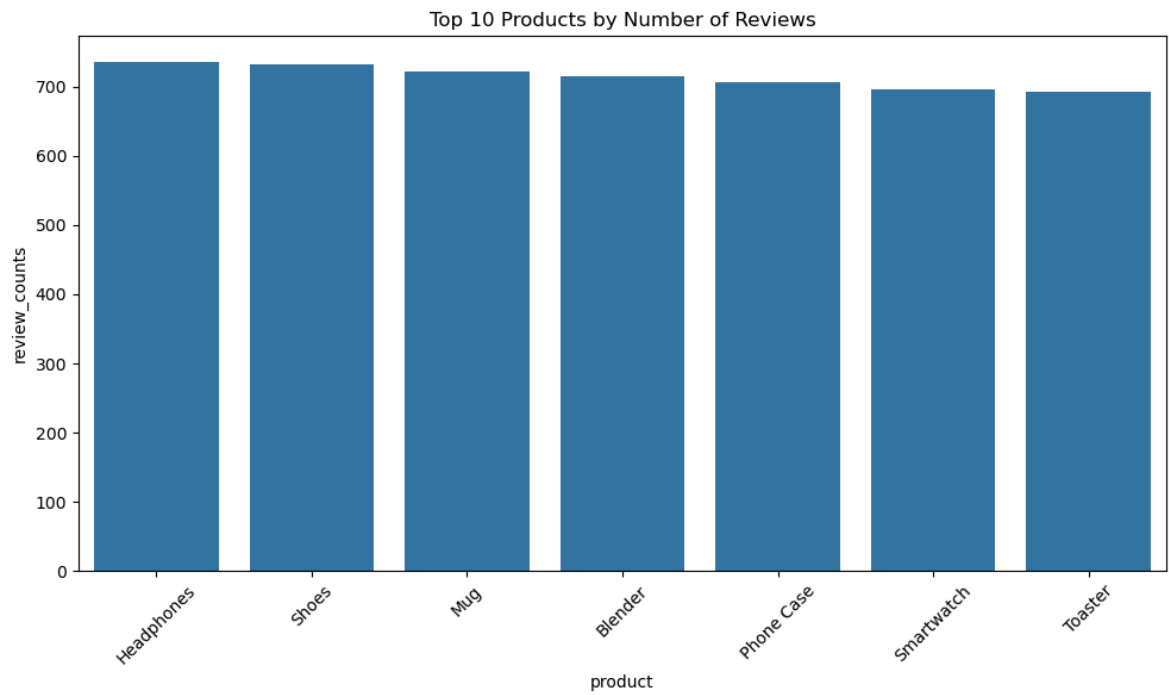
Requirement already satisfied: tzdata>=2022.7 in c:\users\nancy\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2023.3)

Requirement already satisfied: six>=1.5 in c:\users\nancy\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.16.0)

Note: you may need to restart the kernel to use updated packages.

```
In [28]: import matplotlib.pyplot as plt
import seaborn as sns
review_counts = df["product"].value_counts().head(10)
# Show top 10 for clarity
```

```
In [34]: plt.figure(figsize=(10,6))
sns.barplot(x=review_counts.index, y=review_counts.values)
plt.xticks(rotation=45)
plt.xlabel("product")
plt.ylabel("review_counts")
plt.title("Top 10 Products by Number of Reviews")
plt.tight_layout()
plt.show()
```



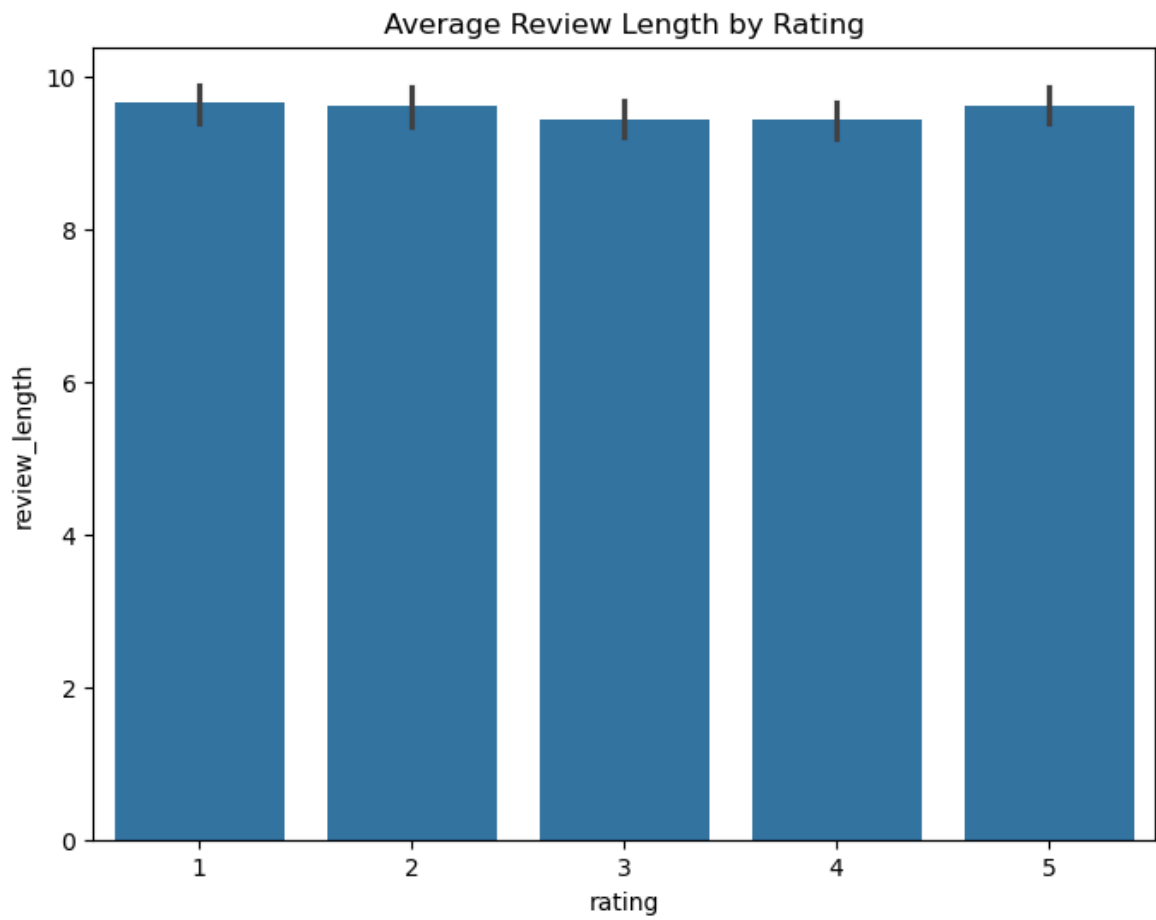
13. Do longer reviews tend to be more positive or more negative based on the review_length and rating relationship?

```
In [33]: plt.figure(figsize=(8,6))
sns.boxplot(data=df[df['sentiment'].isin(['Positive', 'Negative'])], x='sentimen
plt.title("Review Length vs Sentiment")
plt.show()
```



or

```
In [32]: plt.figure(figsize=(8,6))  
sns.barplot(data=df, x='rating', y='review_length')  
plt.title("Average Review Length by Rating")  
plt.show()
```



In []: